MIL-M-36314B
9 September 1976
SUPERSEDING
MIL-M-36314 (DM)
28 October 1964 and
MIL-M-0036314A (DSA-DM)
9 January 1973

MILITARY SPECIFICATION

MICROTOME, Rotary, Automatic

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of automatic-feed, rotary microtome, suitable for serial sectioning of specimens.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Boxes, Shipping, Fiberboard.

Military

MIL-P-116	Preservation-packaging, Methods of.
MIL-M-7298	Manual, Technical: Commercial Equipment.
MIL-K-36311	Knives, Microtome.
MIL-O-36312	Object Disks, Microtome.

FSC 6640

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:23Z
Check the source to verify that this is the current version before use.

STANDARDS

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Material shall be suitable for the purpose intended and as herein specified.

* 3.2 Construction. Microtome shall be the rotary type, and shall consist of a base, object drive, feed mechanism, knife holder and cover. Except for the terminal components (drive wheel and slide block), the horizontal feed mechanism shall be entirely independent of the vertical movement of the object drive and shall be built into the base casting.

3.2.1 Base. Base shall be gray cast iron and shall furnish support for the knife holder, object drive and feed mechanism.

* 3.2.2 Object drive. Object drive shall be direct and shall consist of a hand (drive) wheel, shaft and slide block and locking mechanism for hand wheel. Wheel shall be mounted on the right side of the microtome (knife holder proximal to the user). Movement of the slide block shall be vertical. Each clockwise rotation of the drive wheel shall cause one excursion of the slide block. When activated, locking mechanism shall prevent rotation of the drive wheel.

3.2.3 Feed mechanism. Feed mechanism shall enable the microtome to cut specimens of desired thickness by correlating movement of the hand wheel and slide block with the indicator (3.2.4) and shall consist of a drive wheel, eccentric cam, feed pawl, ratchet, connecting rods and slide block. Clockwise rotation of the hand wheel shall move the slide block forward toward the knife holder. Ratchet wheel shall be the connecting point between the indicator and feed mechanism. Ratchet wheel shall be notched with each notch representing 1 micron of forward movement of the slide block. Feed pawl shall disengage from the ratchet wheel when the hand wheel is turned in a counterclockwise direction or the slide block has moved forward 28 millimeters (mm.). Means shall be provided to reverse the feed mechanism so that the slide block may be moved to its original position.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:23Z
Check the source to verify that this is the current version before use.

3.2.4 Indicator. Indicator shall consist of a conveniently placed knob, a micrometer screw and an easily visible scale. Knob shall turn the micrometer screw and the scale. Gearing between the micrometer screw and the ratchet wheel shall be such that specimens may be set to cut at any desired thickness from 1 to 50 microns. Scale shall be permanently marked from 0 to 50 in 1 micron intervals; zero and succeeding fifth intervals shall be numbered.

3.2.5 Knife holder. Major portions of the knife holder shall be fabricated of corrosion-resisting steel and shall furnish rigid stress for clamping of the knife. There shall be two clamps, each clamp at least 1/2-inch wide and so placed that they do not interfere with the cutting of specimens. The holder shall contain devices which make it possible to adjust the position of the knife while the knife is secured in the holder. Forward, backward, lateral, and tilt adjustments shall be possible. Each adjustment shall be independent and the position of the knife edge shall not change, other than in the direction of the adjustment being made. Reference scale which is marked to indicate knife tilt in 2-degree intervals shall be placed near the holder. Range of tilt adjustment and graduated portion of reference scale shall be as least 30 degrees.

* 3.2.6 Cover. The cover shall be hinged to the base to permit access to the mechanism.

3.3 Accessories. The following accessories shall be furnished with the microtome.

a. Two knives. Knives shall be in accordance with MIL-K-36311, Size 2 except that one knife may be furnished without a handle (see 6.4).

b. Three object disks. Object disks shall be in accordance with MIL-O-36312, Sizes 1, 2 and 3 (one each) (see 6.4).

c. One safety razor blade holder. Holder shall be fabricated of corrosion-resisting steel and shall be 4.625 ± 0.010 inch long. Holder shall have a clamp 1.4375 ± 0.002 inch long which is centered lengthwise between the ends of the holder. Clamp shall be capable of securely locking an extra-heavy or wafer-thin razor blade in place and shall provide even pressure throughout the length of the razor blade.

d. Three packages of razor blades (10 each). Razor blades shall be extra heavy steel, single-edge razor blades. Blades shall be 0.009 ± 0.0005 inch thick, 1-9/16 ± 5/64 inch long, and 23/32 ± 5/64 inch wide. Hardness, measured at the edge of the blade adjacent to the cutting edge, shall be not less than 795 and not more than 857, using a pyramidal diamond and a 5 kilogram (kg.) load on the Vickers hardness tester. Steel used in the manufacture of the blades shall have a carbon

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:23Z
Check the source to verify that this is the current version before use.

content of not less than 1.15 percent and not greater than 1.35 percent, and shall show a uniform, well-spheroidized cementite structure, with carbide particles of uniform size and distribution, free of laminations and non-metallic inclusions. The surface shall be free of rust and scale.

e. One can of pike oil. Can shall contain at least 3 ounces of oil.

3.4 Performance.

3.4.1 Skipping. The ratchet wheel shall not move when tested as indicated in 4.4.1.1, shall move one tooth when tested as specified in 4.4.1.2, and shall make one complete revolution, plus one tooth, minus zero teeth, when tested as specified in 4.4.1.3.

* 3.4.2 Thickness accuracy. The slide block shall move forward 50 ± 2 microns when tested as specified in 4.4.2.

* 3.5 Dimensions. Microtome shall be 16 ± 1 inch long, 14 ± 1 inch wide and 8-3/4 ± 1/2 inch high.

* 3.6 Weight. Weight (without accessories) shall be not less than 72 pounds.

3.7 Finish. Cover shall be finished with epoxy or acrylic resin which shall be gray or blue.

3.8 Identification marking. Each microtome shall be permanently and legibly marked with the name or registered trademark of the manufacturer.

3.9 Service data. The contractor shall furnish, with each unit, two copies each of a technical manual which contains complete instructions for installation, operation and maintenance and lists of component parts. The manual shall comply with the requirements of MIL-M-7298. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the contracting officer, via the cognizant Government inspector, two preliminary copies of his proposed technical manual, for review and comment by the Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary manuals shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of any required revisions and corrections in the final manual without delaying delivery under the contract or purchase order. In addition, the contractor shall furnish 15 copies of the approved technical manual to the contracting officer for distribution. Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of contract

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:23Z
Check the source to verify that this is the current version before use.

and propose tc with those previously accepted, the requirements for mission of 15 additional copies may be waived upon rcficer. Waiver of submission of 15 additional copies s Government.

3.10 Work all be first class throughout. Microtomes shall be free f III and other defects which detract from their appearan rviceability.

4. QUALITS

4.1 Respon less otherwise specified in the contract or purchase onsible for the performance of all inspection requireme xcept as otherwise specified in the contract or purch y use his own or any other facilities suitable for the pe n requirements specified herein, unless disapproved by vernment reserves the right to perform any of the ins pecification where such inspections are deemed necess services conform to prescribed requirements.

4.1.1 Reco ons and tests performed by or for the contractor shall b tor and made available to the Government, upon the Gove me, or from time to time, during the performance of thi of 3 years after delivery of the supplies to which such

4.1.2 Inspe in this specification, is defined as both examination (s vestigation without the use of special laboratory appl testing (determination by technical means of physical and e item.

4.1.3 Certi ate of quality, supplied by the manufacturer of the razor bl lieu of actual performance of such testing by the contract s been maintained and can be demonstrated to the Governm include the name of the contractor, the contract numbecturer or supplier, the NSN, the item identification, the lot number, the lot size, the sample size, the date , individual test results, and the specification requireme

o-

l

they

ed

r-
aced
le

otome.

6312,

-

lder.
lade
.

a
k,
e
:
ickers
n

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:23Z
Check the source to verify that this is the current version before use.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and Table I. Unit of product shall be one microtome.

Table I For examination

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	100%	*
Minor defects	100%	*
For dimensional examination	S-1	2.5

* Items having defects (see 4.3) shall not be furnished.

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and Table II. Unit of product shall be one microtome. Acceptance number shall be zero.

Table II For tests

Characteristic	Requirement	Test procedure	Inspection level
Skipping	3.4.1	4.4.1.1 4.4.1.2 4.4.1.3	100% of microtomes shall be tested.
Thickness accuracy	3.4.2	4.4.2	S-1

4.3 Examination. The microtomes shall be examined for defects including, but not limited to, those listed in Table III.

Table III Classification of defects

Category	Defect
Major	
101	Component part missing.
102	Service data missing.
103	Microtome not free of sharp edges (excluding knives and razor blades).
104	Microtome not free of rust spots.
105	Applicable components not secured.
Minor	
201	Painted surfaces not free of blisters and cracks.
202	Microtome not free of scratches.
203	Identification marking not complete, correct, legible and permanent.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:23Z
Check the source to verify that this is the current version before use.

4.3.1 Dimensional examination. The microtomes shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

* 4.3.1.1 Packaging inspection. The inspection of the packaging and interior package marking shall be in accordance with the group A and B quality conformance inspection requirements of MIL-P-116. The inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Skipping. The following tests shall be performed.

4.4.1.1 Zero micron setting. Set indicator at zero and rotate hand wheel at least one revolution.

4.4.1.2 One micron setting. Set indicator at 1 micron and rotate hand wheel one revolution.

4.4.1.3 Complete revolution. Divide the number of teeth on the ratchet wheel by 10. Set quotient on the indicator. Rotate the hand wheel 10 revolutions at 3 revolutions per second.

4.4.2 Thickness accuracy. A precision dial gage of known accuracy and graduated in microns shall be secured to the knife holder. Indicator shall be set at 10 microns, and the hand wheel shall be rotated five revolutions. Object disk may be secured to the slide block. Any tolerance on the dial gauge shall be subtracted from the tolerance specified in 3.4.2.

5. PACKAGING.

5.1 Packaging. Packaging shall be Level A or C as specified (see 6.2).

5.1.1 Unit of issue. One microtome, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit preparation. Accessories shall be unit packaged Level A in accordance with unit package requirements as indicated in the applicable specification for the particular item. Blade holder shall be packaged in a carton of appropriate size and design. Each razor blade shall be placed in an individual paper envelope. Ten

Source: https://assist.dla.mil -- Downloaded: 202610-21T08:23Z
Check the source to verify that this is the current version before use.

blades shall be placed in a carton of appropriate size and design. All accessories, including the 3-ounce can of pike oil, shall be overpacked in a carton of appropriate size and design, and identified as to contents and quantity of each accessory.

* 5.1.2.2 Unit packaging. Each unit, with component parts, and accessories packaged as stated above, shall be packaged in a box of appropriate size, constructed in accordance with PPP-B-636, type CF, class weather-resistant, V3c. Box shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of or damage to the contents. Box shall be closed and waterproofed as specified in the appendix of PPP-B-636. Packaging method shall conform to MIL-P-116, method III.

5.1.3 Level C. Units shall be packaged in standard commercial containers of the type, size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be Level A, B or C as specified (see 6.2).

* 5.2.1 Level A. Each unit shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification.

* 5.2.2 Level B. Each unit shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Fiberboard box shall be closed by method II of PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

* 5.3 Marking. Each unit and exterior container shall be marked as specified in MIL-STD-129.

6. NOTES

6.1 Intended use. Microtome is intended for use in cutting specimens for microscopic examination.

Source: https://assist.dla.mil -- Downloaded: 202640-21T08:23Z
Check the source to verify that this is the current version before use.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. National Stock Number (NSN).
c. Selection of applicable levels of packaging and packing (sec 5.1 and 5.2).

6.3 This specification covers the following item appearing in the Federal Supply Catalog:

National Stock No.	Item Identification
6640-00-433-6000	MICROTOME, Rotary, Automatic

6.4 The knives referred to in 3.3a are NSN 6640-00-433-9500, Knife, Microtome, 120 mm.

6.5 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
DSA-DM

Project No. 6640-0968

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:23Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:23Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-M-36314B, Microtome, Rotary Automatic

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:23Z
Check the source to verify that this is the current version before use.

FOLD

Defense Personnel Support Center
2800 South 20th Street
Philadelphia, PA 19101

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

POSTAGE AND FEES PAID
DEFENSE SUPPLY AGENCY
DoD - 304



Commander
Defense Personnel Support Center
ATTN: DPSC-ATT
2800 South 20th Street
Philadelphia, PA 19101

FOLD

Source: https://assist.dla.mil -- Downloaded: 202640-21T08:23Z
Check the source to verify that this is the currentversion before use.